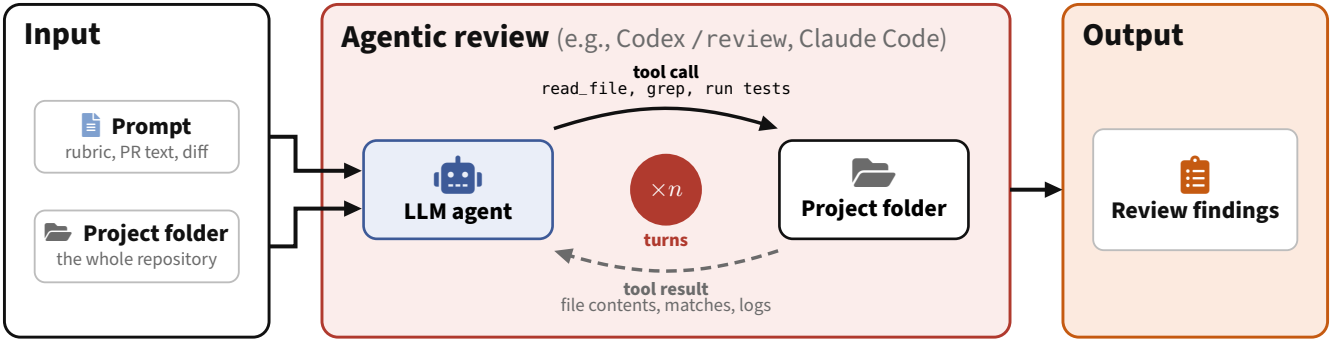
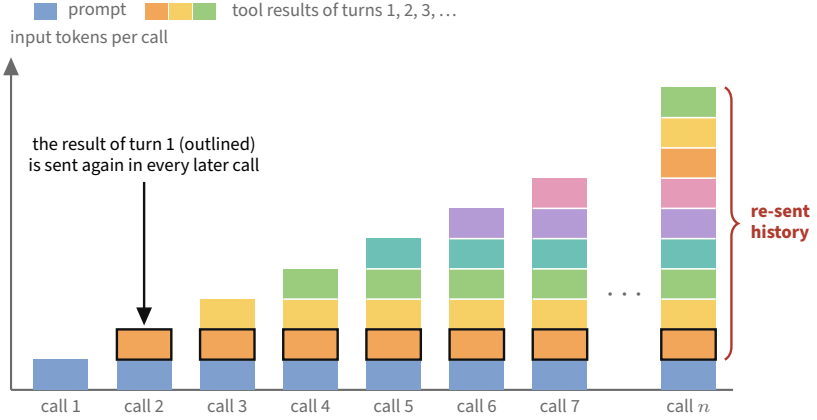


Problem: every turn is a new LLM call that re-sends everything before it



Why it is expensive

Many calls. One LLM call per tool turn, and a review takes many turns.

Repeated tokens. Call k re-reads the prompt and the results of turns $1, \ldots, k-1$.

Total input

$$\sum_{k=1}^{n} \left(P + \sum_{j<k} r_j \right) \approx nP + \tfrac{n^2}{2}\,\bar{r}$$

grows with the square of the number of turns.